HanuAi ML Assessment — Task 2 Findings

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Executive summary

The file contains 1,000 records and 20 fields, covering opening dates from 2019-08-27 through 2021-06-01. There are 0 exact duplicate rows and 0 duplicate event IDs.
Audio/entertainment/navigation complaints account for 794 records across the three largest labels. The dominant coded causal label is “Module/Component-No/Incorrect Communication” (377).

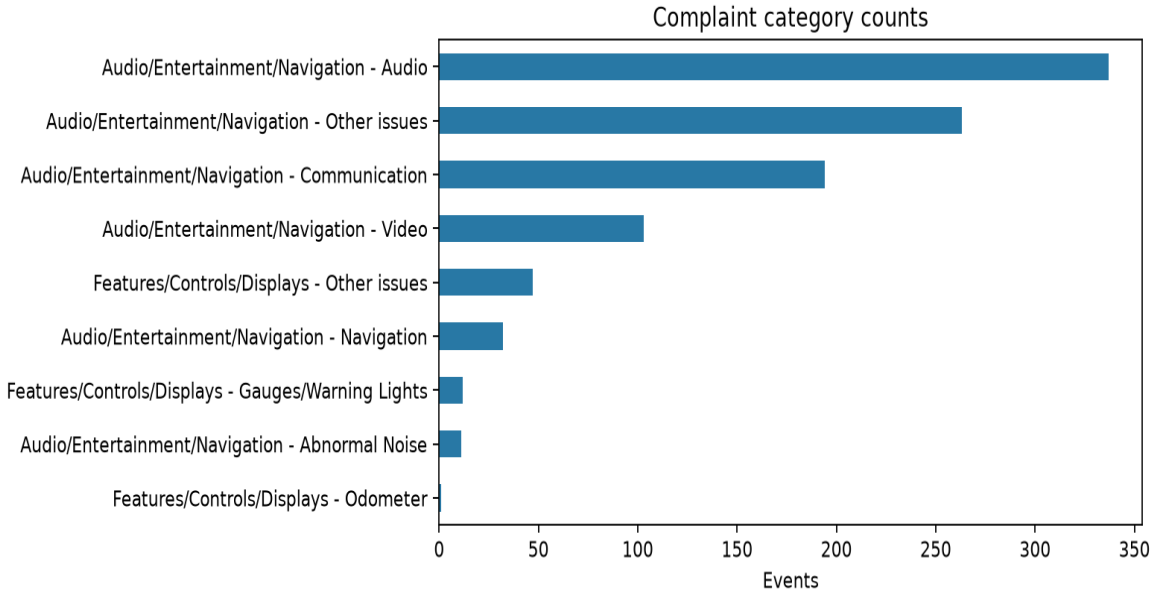
Dataset and data quality

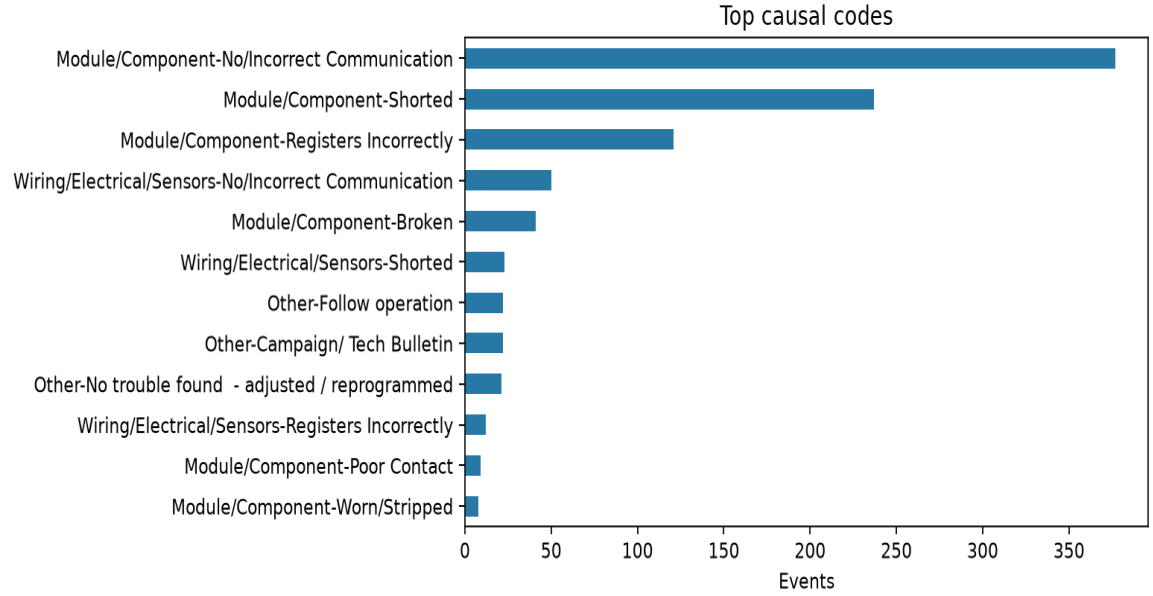
- Missing values are concentrated in the six provided mined-entity fields: each contains values in only 5 of 1,000 rows (99.5% missing). Those labels cannot represent the complete dataset.
- One CAUSAL_VERBATIM and one IN_USE_DATE value are missing; three CAUSAL_CD_DESC values are missing. All 1,000 opening dates parse.
- The provided derived categorical labels have been retained unchanged. A separate cleaned narrative field and separately generated tags are included in events_enriched.csv.
- The greatest monthly count is 69 in 2020-02; monthly counts are descriptive and should not be read as rates because population/exposure denominators are not provided.

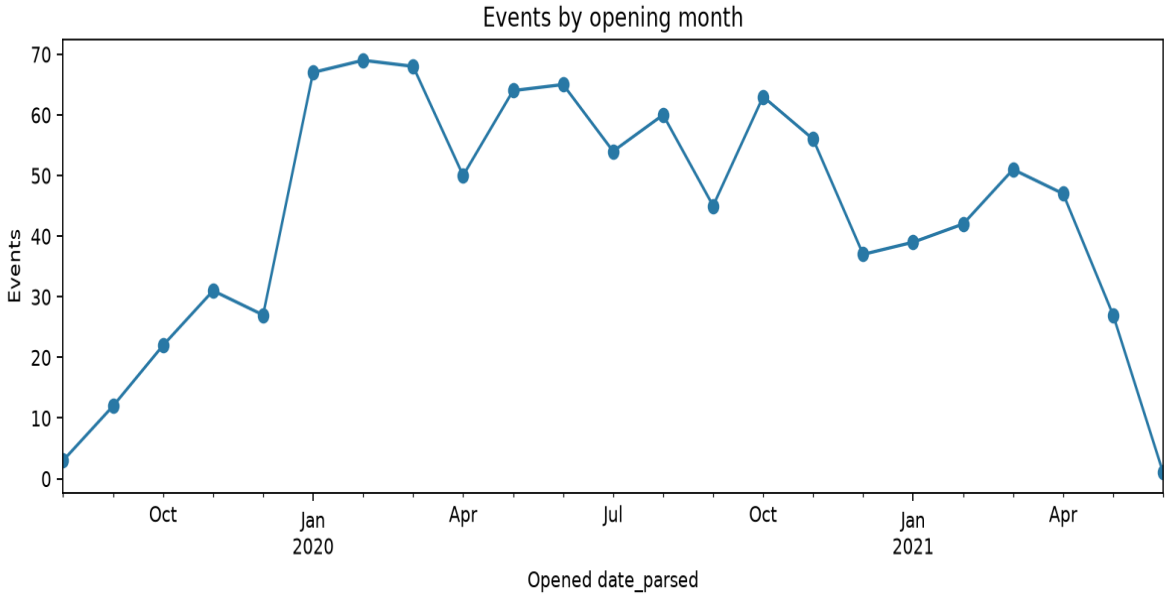
EDA and text mining

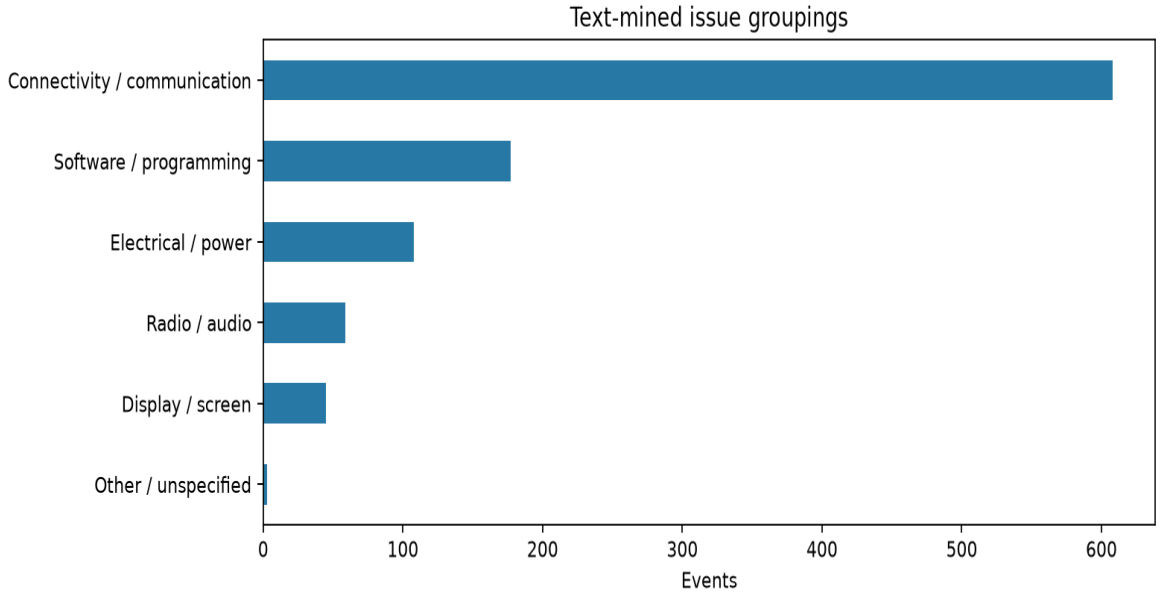
The most common complaint labels are Audio (337), Other issues (263), and Communication (194). The top coded causes include incorrect communication (377), shorted module/component (237), and incorrectly registering module/component (121).

Narrative keyword extraction assigns each row to a primary grouping: Connectivity / communication: 608, Software / programming: 177, Electrical / power: 108, Radio / audio: 59, Display / screen: 45, Other / unspecified: 3. Because categories use transparent overlapping keywords and an explicit priority order, they are an exploratory index for review, not independent diagnosis. Five TF-IDF/K-means clusters are included with top terms in cluster_summary.csv; unsupervised clusters require engineering review before being interpreted as failure mechanisms.


Complaint category counts
Audio/Entertainment/Navigation - Audio
Audio/Entertainment/Navigation - Other issues
Audio/Entertainment/Navigation - Communication
Audio/Entertainment/Navigation - Video
Features/Controls/Displays - Other issues
Audio/Entertainment/Navigation - Navigation
Features/Controls/Displays - Gauges/Warning Lights
Audio/Entertainment/Navigation - Abnormal Noise
Features/Controls/Displays - Odometer
0
50
100
150
200
250
300
350
Events


Top causal codes
Module/Component-No/Incorrect Communication
Module/Component-Shorted
Module/Component-Registers Incorrectly
Wiring/Electrical/Sensors-No/Incorrect Communication
Module/Component-Broken
Wiring/Electrical/Sensors-Shorted
Other-Follow operation
Other-Campaign/ Tech Bulletin
Other-No trouble found - adjusted / reprogrammed
Wiring/Electrical/Sensors-Registers Incorrectly
Module/Component-Poor Contact
Module/Component-Worn/Stripped
0
50
100
150
200
250
300
350
Events





Recommendations

1. Prioritize a joint review of radio/audio communication events, module communication DTCs, and repair narratives to determine whether common software, module, or network conditions explain repeat visits.
2. Audit the 995-row gap in the supplied entity labels and backfill extraction using the same field definitions before using tags for operational dashboards.
3. Compare rates by vehicle/build cohort only after joining production and in-service denominators; the present event counts cannot identify relative risk.
4. Validate extracted tags and cluster labels with domain reviewers, then track precision/coverage before automating triage.

5. Preserve event-level traceability and investigate cases with no trouble found or reprogramming alongside confirmed part replacements.

Limitations

The file is a 1,000-row sample with no exposure denominator and only a single model year (MODLYR=2020). Correlation in coded labels and text does not prove root cause. Keyword extraction can miss paraphrases and negation. The six supplied entity fields are sparsely populated. Date trends may reflect selection or reporting practices.

Reproducibility

Run python -m src.task2 --input data/raw/Hanuai_task2_assignment.csv --output outputs/task2, then python -m src.build_report. Outputs include the row-level enriched CSV, profile, summary JSON, monthly counts, cluster terms, and charts.